

of Louis XIV's wars were aggressive and designed to produce a rapid result. These tactics involved digging trenches towards enemy fortifications in a zig-zag, so that fire could not rake the trench from end to end. At intervals, engineers dug transverse "parallels", lines to which the siege cannon then advanced. Under Vauban's system, the third parallel was the last. From there, soldiers might emerge from their trenches, with grenadiers at the fore, to mount an assault on the "covered way" – the defensive position held by the enemy infantry, on the outer edge of the fortress's ditch. Once this position was taken, siege cannon could advance close to the walls, and the defenders would be expected to surrender – a gesture rewarded by honourable treatment. Of course, actual sieges only approximated this pattern: for example, tunnelling sappers might blow up walls using mines, or moats might be drained or crossed by soldiers on rafts.

Ironically, despite the money lavished upon them, by the 18th century fortifications rarely resisted a siege for long. The golden age of siege warfare and fortifications was over by 1720.

"The more powder we burn, the less blood we lose"

ATTRIBUTED TO MARQUIS DE VAUBAN, AT THE SIEGE OF CHARLEROI, 1693



◀ THE SACK OF MAGDEBURG

If defenders refused to surrender, they faced massacre when fortifications were taken by assault. At Magdeburg, in 1631, over 20,000 people were killed when the city fell after a six-month siege.



◀ THE SIEGE OF MAASTRICHT

In June 1673, French King Louis XIV, invading the Netherlands, besieged the fortress city of Maastricht with an army of 45,000 men. Vauban directed the siege, overcoming the defences within the month.

KEY EVENTS

1650–1700

- **1667** The French army details specific soldiers to specialize in throwing grenades – the first instance of grenadiers.
- **1674** Dutch military engineer Menno van Coehoorn introduces a man-portable mortar gun for use in sieges.
- **1678** French King Louis XIV appoints Vauban his commissioner-general of fortifications, embarking on a large-scale programme of fortress-building around the frontiers of France.
- **1685** Van Coehoorn publishes his influential treatise entitled *New Fortress Construction*.
- **1692** Vauban conducts the Siege of Namur (see pp.180–81), in which van Coehoorn directs the defence; the fortress falls to the French in five weeks.
- **1695** After a two-month siege conducted by van Coehoorn, the Grand Alliance retakes Namur from the French.